

Chapter 25

Capturing From the Cintel Film Scanner

This chapter details how to ingest scanned film using DaVinci Resolve settings and workflows to control the Cintel film scanner.

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Controlling the Cintel Film Scanner

The Blackmagic Cintel film scanner is a compact, easy to use, real time film scanner capable of converting 35mm, 16mm, and 8mm (with separately purchased gates) positive and negative film formats into Cintel Raw Image (CRI) digital files that can be organized, edited, and graded using DaVinci Resolve, delivered to any format DaVinci Resolve can output, and archived for later use.



The Cintel scanner

DaVinci Resolve can control any Blackmagic Cintel film scanner that's connected to your computer via Thunderbolt or PCIe. Once connected, the Film Scanner controls in the Media page can be enabled, which let you choose the film type to be scanned, align the film frames to the sensor, adjust the scanner's light source for optimal exposure and color, and choose whether to use the scanner's hardware-based Automatic Perf Detection to perform image stabilization.

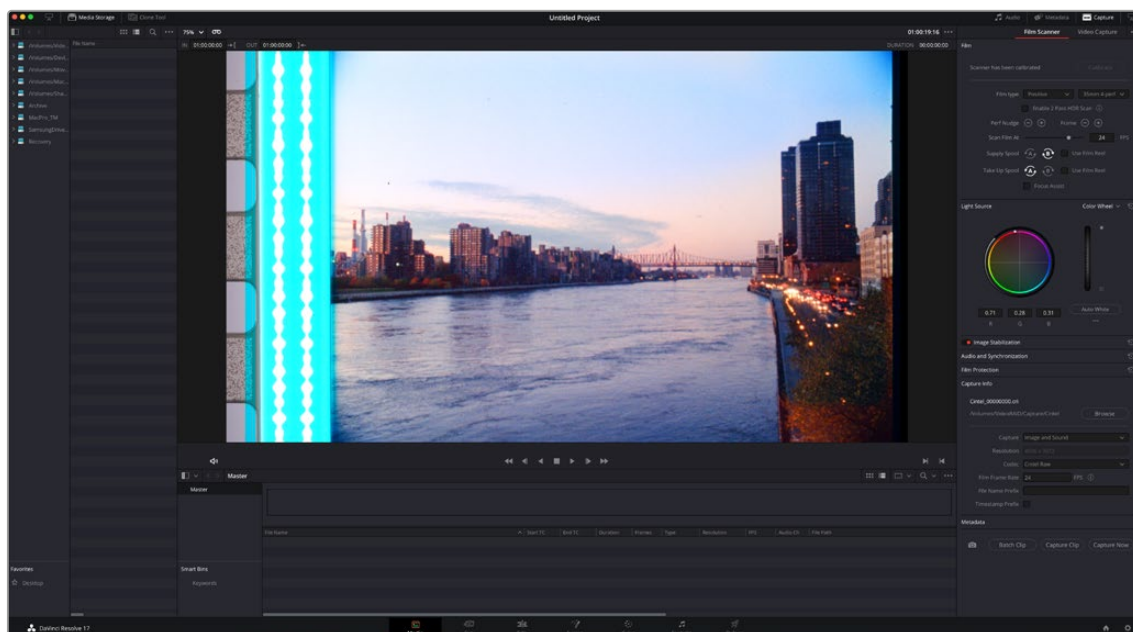
NOTE: This chapter of the DaVinci Resolve manual describes the use of a Cintel film scanner connected to DaVinci Resolve for the purpose of ingesting scanned film. For other operational inquiries, please see the documentation that accompanies the scanner itself, or visit the Blackmagic Design support page on the web to download it.

This section of the manual shows you how to use settings and features in DaVinci Resolve's film scanner panel to control your scanner. For example calibrating your scanner, adjusting the light source strength and color temperature, setting image stabilization, and more. You can even set how gentle your Cintel scanner handles film which may have become delicate with age.

TIP: DaVinci Resolve saves all scanner settings in your current project.

The Cintel Scanner Interface

Click on the 'capture' button in the UI toolbar at the top of the DaVinci Resolve screen to set the media page to control your Cintel scanner. Open DaVinci Resolve's film scanner panel to set up, calibrate, and choose options for logging or scanning a selected range of the currently spooled roll of film. If you want more room for viewing the Cintel scanner controls, click the full height button that's all the way to the right of the UI toolbar, and turn off the 'metadata' panel.



Cintel scanner controls in the Media page

Transport Controls: The transport controls under the viewer, while similar in appearance to those used while in playback mode, now work to control the Cintel scanner. Additional controls appear for moving forward or backward a frame at a time.

In and Out Controls: In Cintel Scanner mode, the In and Out buttons to the right of the transport controls define a range of the film roll from which to capture.

The following groups of settings appear to the right of the Media page viewer when in Cintel Scanner mode to scan clips from film into the Media Pool.

Film Controls

The film controls provide options to calibrate the scanner's optics, select the film type, adjust the frame alignment, scan speed and spool wind direction, choose the reel type, and turn the focus assist feature on or off.

Calibration

This option lets you calibrate the optics of the scanner to eliminate optical blemishes or dust that cannot be removed. Please note that this feature does not remove dust from the film itself.

Scanner has not been calibrated

Calibrate

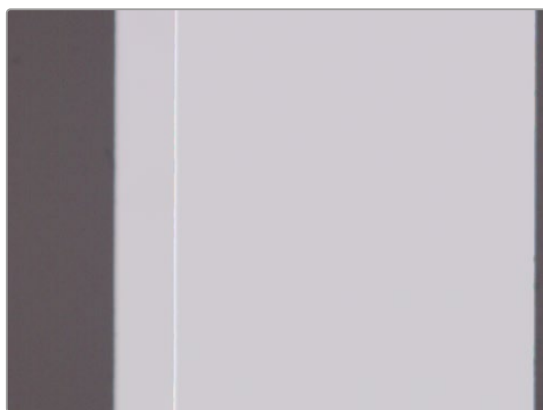
The calibration button can be used to help remove dust or small blemishes from the optics of your Cintel scanner.

While it's recommended to "spray dust" the optics before scanning new material, it's possible over time for some blemishes on the optics to be unremovable, in which case using the calibrate button will eliminate them from the scanned image.

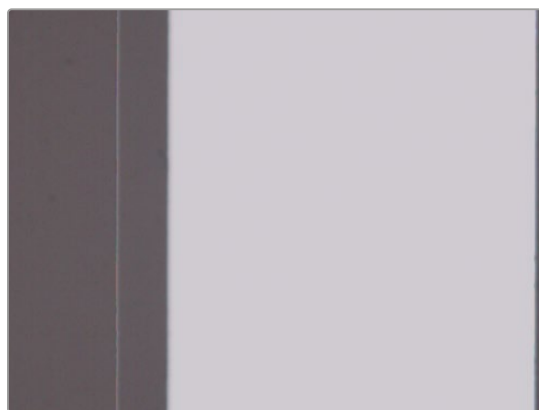
Calibrate the optics with the skid plate installed and correctly aligned, as this assists with image stabilization and offers the best image quality.

To support small gauge film types, the calibration button also analyzes the gate fitted and adjusts the captured image to a datum reference. To permit this, the calibration process should be run on every gate swap with no film in the gate to ensure optimum performance.

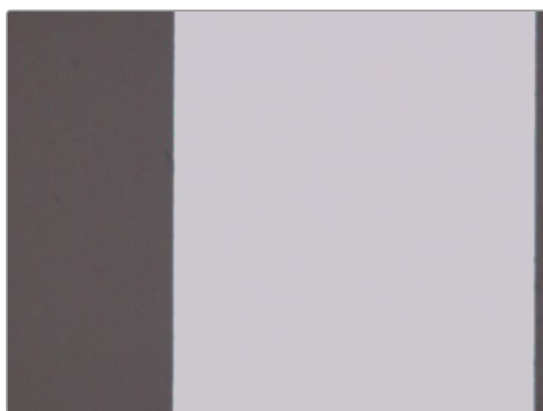
If you are using the 8mm HDR gate or 16mm HDR Gate and have adjusted the aperture shutters to switch to a different film gauge, you will need to repeat the calibration process to avoid a 'ghost image' of the previous calibration.



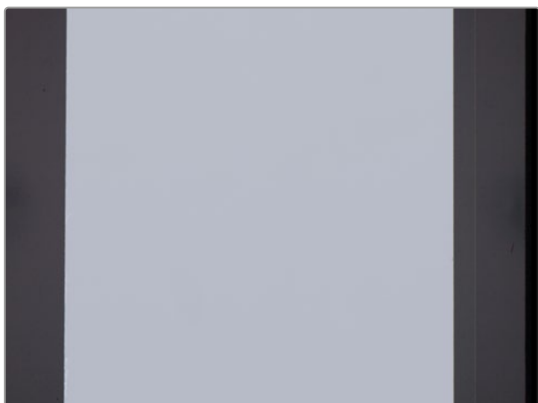
Incorrect calibration shows Super 8mm film with a Standard 8mm calibration



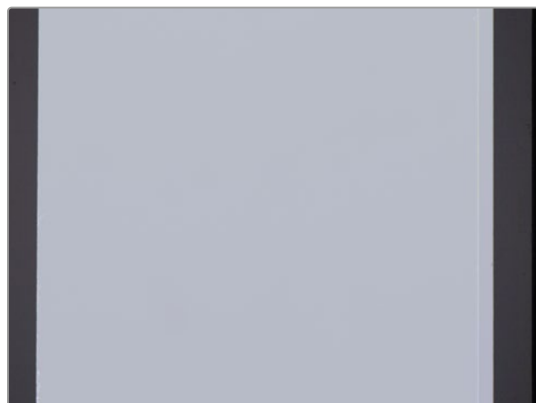
Incorrect calibration shows Standard 8mm film with a Super 8mm calibration



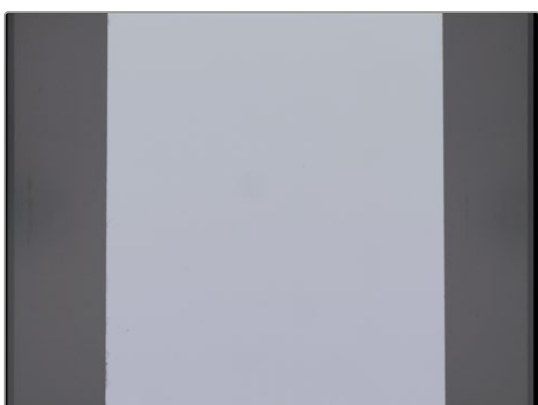
Correct calibration of Standard 8mm film



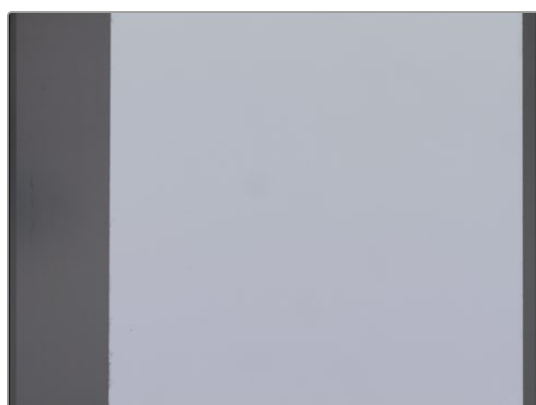
Incorrect



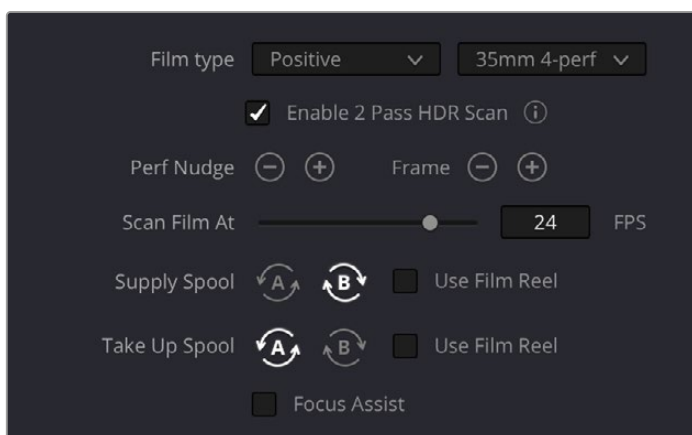
Incorrect



Super 16mm poss correct



Standard 16mm neg correct



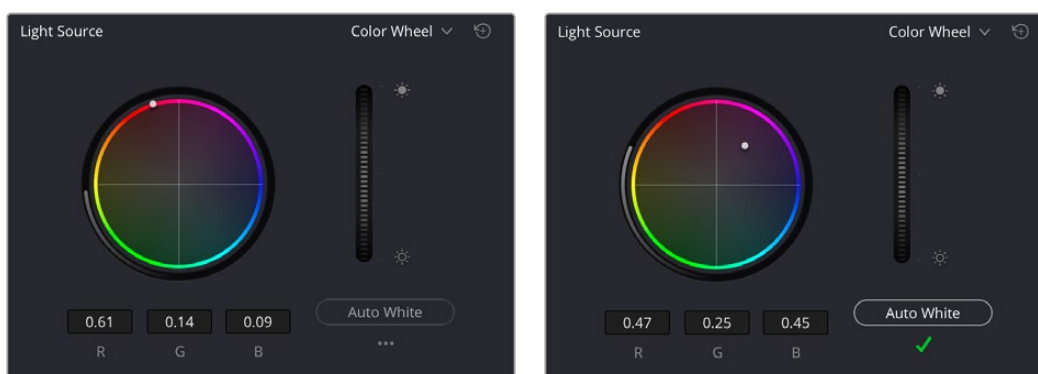
Film Type controls in the Media page

Film Type

Lets you choose what type of film you're scanning. For color as well as black and white film, the choices are positive, negative, interpositive, and internegative. HDR scanning offers an improvement for all these film types. Select the film type you're scanning from 35mm 2, 3, and 4 perf, 16mm, Super 8mm, and 8mm. The scanner automatically detects whether the film is 35mm, 16mm, or 8mm. For 8mm scanning, the default is set to Super 8mm. When scanning regular 8mm, you will need to manually set your scanner for that specific film type.

When scanning interpositive and internegative film on Cintel Scanner and Cintel Scanner G2, the increased density of the film requires slightly extended pulse durations from the light source. Normally, this does not affect the scan, however, a slight reduction in resolution may occur when scanning at above 12 frames per second. If you do notice a difference in resolution, simply reduce your scanning speed to 12 frames per second or less.

- **Enable 2 Pass HDR Scan:** Enables high dynamic range multi pass capture. It is important to perform an 'auto black' or 'auto white' on a frame with a wide dynamic range as it determines the high and normal exposure levels from your selected frame.
- **Perf nudge:** Used for making fine adjustments of the perf position relative to the scanner gate aperture. Command-J nudges up, while Command-L nudges down.
- **Frame:** These buttons are push and hold to activate. When on, the film is slowly advanced to move the frame up or down and when released the film stops in place. This is useful for aligning the film frame with the scanner's sensor. Using the 'perf nudge' and 'frame' buttons, you want to align the visible film frame so the bottom of the previous frame and the top of the next frame are just visible at the top and bottom of the viewer, and the current frame is centered vertically.
- It's important to make sure the image in the viewer is not zoomed in when you do this. Command-Left Arrow on your keyboard moves the frame up, while Command-Right Arrow moves the frame down.
- **Scan Speed:** With adequate disk performance, you should be able to scan at 30 fps. However, if you're scanning to a slow hard drive, you can reduce the scanning speed to a frame rate that's suitable for your workstation without dropping frames.
- **Supply:** Sets the wind direction of the left-hand side feed spool. While auto-detection will prevent incorrect operation, you should manually configure the reel winding direction based on how each film roll is wound.
- **Take up:** Sets the wind direction of the right-hand side take up spool. While auto-detection will prevent incorrect operation, you should manually configure the reel winding direction based on how each film roll is wound.
- **Use Film Reel:** Small film reels have a different weight and inertia compared to large film spools, and this can affect the transport system. Tick this box to switch to settings that offer improved stability for small film reels.
- **Focus Assist:** Enables luminance peaking on your scanner's HDMI monitor output, plus the viewer inside DaVinci Resolve's film scanner panel, which makes it easy to obtain optimum focus adjustments.



Light Source controls in the Media page showing the default uncalibrated status of the light source (left), and the status when calibration is successful (right)

Adjusting the Light Source

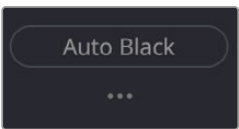
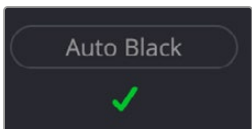
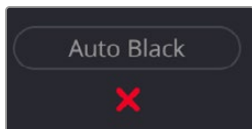
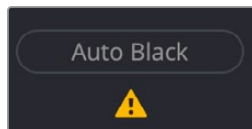
These controls let you adjust your Cintel Scanner's light source to calibrate the optimal Dmin density minimum. The Dmin density minimum is the minimum scanned value, plus the color temperature of the scanned material. Adjusting the light source settings correctly will ensure the best quality scans and make sure you are not clipping image data during the scanning process. It's a good idea to check your light source settings when changing film to ensure the quality of your capture.

Use the built in software scopes in DaVinci Resolve to help you set your light source to its optimal level settings. The scopes can be opened in the Media page by choosing Workspace > Video Scopes > On.

- **Light Source master wheel:** The vertical light source master wheel is located next to the color wheel and adjusts the intensity of the light source used to illuminate the film, raising or lowering the RGB channels all at once. For typical negative film, this lets you adjust the black point of the film image, which is the darkest part of the image. In negative film, this in fact corresponds to the highlights of the film image. Adjust the light intensity to sit just above the typical Dmin value of 95, as measured on the histogram of the video scopes, which guarantees that the highlights won't be clipped by a Cineon-style LOG conversion. For positive film, simply adjust the master wheel so that no part of the signal is being clipped.
- **Auto Black and Auto White button:** Analyzes the current frame displayed in the Viewer and does an automatic adjustment to set the black point for negative and internegative using the framing bar area. For print and interpositive film it uses the brightest highlight in the image area to set the white point, so key selection of the frame is important. Alternatively, for positive film types you can use a punch hole frame for maximum white to ensure all subsequent frames are compliant. For positive film types, the 'auto black' button changes to 'auto white'.

Light Source Status Indicators

These indicators under the 'auto' button let you know if the auto black or auto white light source calibration has been successful or if there are items to address. They also provide a helpful reminder to recalibrate the LED light source to ensure the highest quality scans.

 Light source calibration has not yet been performed.	 Light source calibration has performed successfully.	 Light source calibration failed. Verify you have the correct film type selected, the light path is not obstructed and a suitable reference frame is selected. If required, try advancing film and selecting an alternate reference frame.	 Light source calibration has successfully balanced colours but not at desired levels. As a solution, you can accept the balanced scan and optimize the image in DaVinci Resolve's color page or select an alternate reference frame for calibration with a wider dynamic range.
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Light Source Status Indicators

These indicators under the 'auto' button let you know if the auto black or auto white light source calibration has been successful or if there are items to address. They also provide a helpful reminder to recalibrate the LED light source to ensure the highest quality scans.

- **RGB controls:** By default, a color balance control lets you adjust all three color channels by varying amounts to alter the color temperature of the light source used to illuminate the film,

while the adjusted R, G, and B values are displayed in three fields below. Optionally, you can choose to put this control into 'color bars' mode using the mode pop-up to the right of the 'light source' title bar, which changes this control to three vertical red, green, and blue color channel sliders. Image Stabilization

NOTE: The light source calibration is saved when you change rolls. This allows you to scan multiple rolls with a single light source calibration in both SDR and HDR modes. The last light source calibration will be saved until a new calibration is performed, or until your Cintel scanner is power cycled.

These controls let you enable and disable as well as control image stabilization to eliminate vertical film hop and horizontal weave

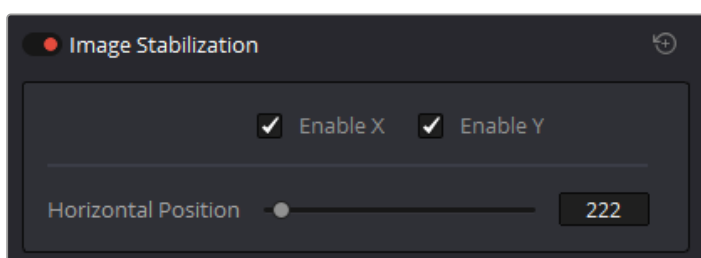


Image Stabilization controls in the Media page

- **Image Stabilization enable/disable control:** The dot to the left of the 'image stabilization' title bar lets you enable or disable your scanner's hardware-based image stabilization altogether. While hardware stabilization is typically desirable when you have high quality perforations, you may want to turn this option off if the condition of the perforations is poor and you decide to use DaVinci's software based stabilization instead.

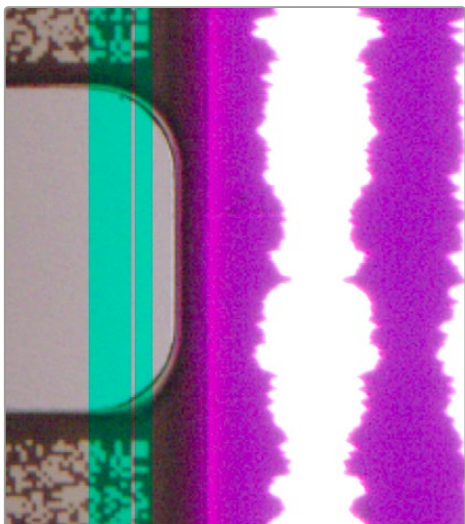
When image stabilization is enabled, a horizontal X axis detection overlay is displayed in the viewer, highlighting the edge of the film perforation that will be used as the reference for stabilization. This overlay is automatically hidden when recording. Image stabilization is enabled by default.

- **Enable X and Y checkboxes:** Enable X and enable Y lets you choose whether to use hardware image stabilization to fix horizontal gate weave and vertical gate hop respectively. If the results are unsatisfactory with both axes enabled, you can turn off the axis that's causing issues with stabilization and utilize DaVinci Resolve's software-based stabilization tools instead.
- **Horizontal Position slider:** Your Cintel scanner attempts to automatically place the stabilization detection overlay at the best location, with reference to the perforation shown on the currently loaded frame, for the best stabilization result.

You will notice a thin transparent line in the blue alignment overlay. For optimum stabilization, this line should touch the edge of the perforation. If the automatic positioning is not ideal, you can manually move the overlay to a more ideal position, either by dragging it in the viewer with your mouse, or by using the horizontal slider.

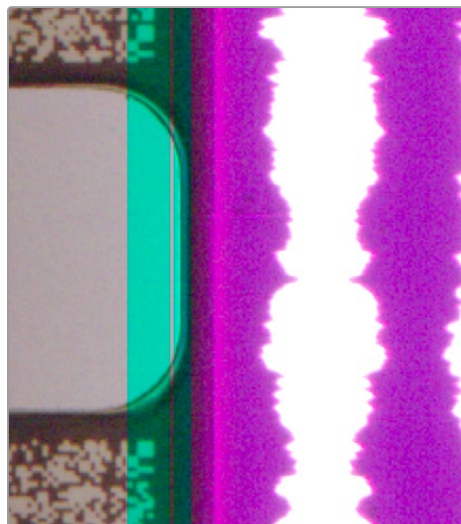
Ideal placement of the stabilization overlay should position the clear line in the alignment overlay on the edge of the perforation, as shown in the example image. With the overlay correctly positioned, this enables hardware stabilization of gate weave along the X axis.

Image stabilization automatically manages vertical gate hop when you select the 'enable y' checkbox. It needs no further adjustment and works in conjunction with horizontal stabilization.



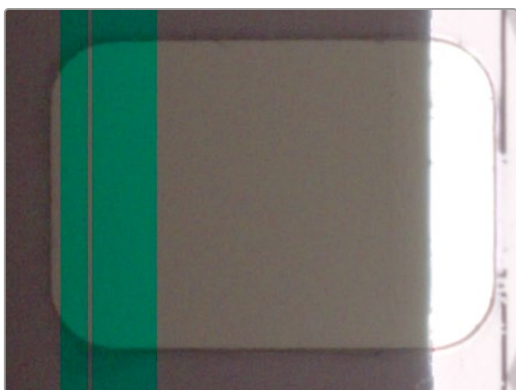
Incorrect 35mm setup.

Adjusting the horizontal position of the stabilization overlay. The overlay is not aligned to the edge of the perforation.



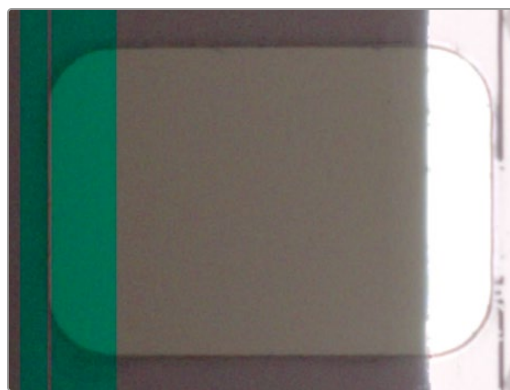
Correct 35mm setup.

The hardware stabilization control correctly positioned over a perforation in the viewer. The transparent stripe in the stabilization overlay touches the edge of the perforation.



Incorrect 16mm perf setup.

The overlay is not aligned to the edge of the perforation. With the 16mm HDR skid plate installed, the stabilizer aligns automatically to the other side of the perforation to avoid interfering with the film image and improve horizontal stability. When using a 'non-HDR' 16mm skid plate, it functions the same as the 35mm skid plate.



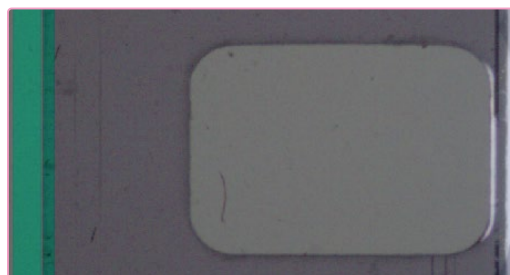
Correct 16mm perf setup.

For the stabilization overlay, the default alignment position is for the left edge of the perforation as this is not affected by image content. The transparent stripe in the stabilization overlay touches the edge of the perforation correctly.



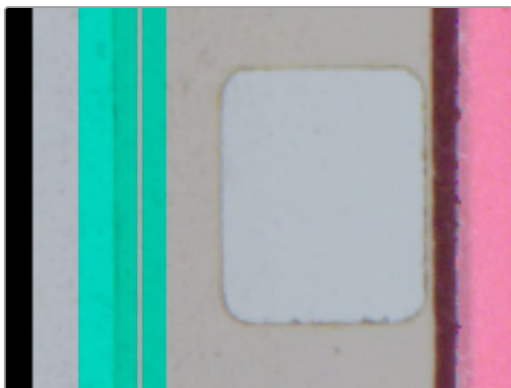
Incorrect 16mm film edge setup.

The overlay is not aligned to the edge of the film reference edge.



Correct 16mm film edge setup.

The overlay is aligned to the edge of the film reference edge. The transparent overlay touches the film ref edge correctly.



Incorrect 8mm setup.

The overlay is not aligned to the edge of the film.



Correct 8mm setup.

For the stabilization overlay, the transparent stripe in the overlay is within the film base area.



Incorrect 8mm perf setup.

The overlay is not aligned to the edge of the perforation.



Correct 8mm perf setup.

For the stabilization overlay, the default alignment position is for the left edge of the perforation as this is not affected by image content. The transparent stripe in the stabilization overlay touches the edge of the perforation correctly.



Incorrect 8mm film edge setup.

The overlay is not aligned to the edge of the film reference edge. The overlay is not aligned to the edge of the film reference edge.



Correct 8mm film edge setup.

The overlay is aligned to the edge of the film reference edge. The transparent overlay touches the film ref edge correctly.

To closely check the results of your stabilization settings before capturing, set the Viewer to full resolution and zoom into the perforation region to allow fine adjustment. Simply click on the options settings at the top right corner of the Viewer and select 'full resolution preview' from the menu,

click on the zoom scalar at the top left-hand corner of the Viewer and select '100%'. This setting does not affect the stabilization feature but enables the best possible preview so you can monitor how well it is performing.

Full resolution preview is very GPU intensive and may result in some frame lag. For best performance, turn full resolution off after checking stabilization. It is worth mentioning that the overlay position will be saved within the project settings and not the clip settings, therefore may require individual adjustment for multiple rolls.

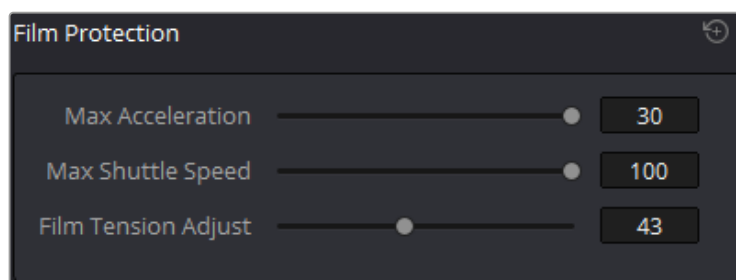
Resetting the detection overlay to its default position for a given film type can be achieved via the Stabilization pane Reset button.

Audio and Synchronization

The Audio and Synchronization palette contains settings for the optional Cintel Audio and KeyCode Reader. Refer to the 'optional Audio and KeyCode Reader' section of the *Blackmagic Cintel manual* for more information.

Film Protection

These controls are intended to allow delicate film to be handled gently by the Cintel Scanner. Fast acceleration and shuttle speeds can be hard on archival footage, so it's recommended to lower both of these sliders from their defaults whenever you're scanning older film.



The 'Acceleration' and 'Shuttle Speed' sliders should be lowered when scanning older, delicate archival film.

NOTE: The 'Film Tension Adjust' slider is present when using Cintel scanners with sprockets.

- **Max Acceleration:** Sets the maximum change in speed to increase or decrease by 5-30 fps per second.
- **Max Shuttle Speed:** Changes the speed of shuttling from one section of film to another between 1–100 frames per second for 35mm film, and between 1–200 frames per second for 16mm film.
- **Film Tension Adjust:** If your Cintel Scanner has sprocket wheels, this setting gives you the ability to adjust the amount of tension applied to 35mm film. For example, when loading delicate archival film, or compensating for film shrinkage. There is no possible way you can damage the film with the 'film tension adjust' setting. The adjustment values are very small and only gentle changes are all that's required to prevent sprocket picking.

If your Cintel Scanner has capstans, sprocket picking cannot occur, so this setting is disabled or not present.

Editing Capture Info Metadata

When DaVinci Resolve is used in conjunction with Cintel Scanner, a set of capture metadata fields appears at the bottom of DaVinci Resolve's film scanner panel. The 'capture info' panel has editable metadata fields that describe capture properties such as where to save files, the type of codec, frame rate, and the format of file names. This metadata is attached to your clips and can be read on the media page.

Before you begin scanning, you may want to adjust some of the project settings.

The 'capture info' panel lets you specify metadata for your scanned clips.

- **Capture Location:** Before you begin a film scanning session, scroll down to the 'capture info' section of DaVinci Resolve's film scanner panel to make sure the scanned files are being saved to the directory and volume where you want them. Click the 'browse' button and choose a location from the file destination dialog. It's good to do this first, as this step is easy to forget.
- **Capture:** When you have a Cintel Audio and KeyCode Reader fitted, this menu gives you options for 'audio only' so no images will be captured, or 'image and sound'. Alternatively, you can capture 'image only' if audio is not important.
- **Resolution:** The resolution of the capture files depends on the source film format so this field cannot be edited.
- **Codec:** DaVinci Resolve selects the 'Cintel Raw' codec for lossless compression by default, or you can choose 'Cintel Raw 3:1' for even smaller file sizes.

- **Film Frame Rate:** Specify the frame rate that the film was originally shot at. DaVinci Resolve automatically adjusts the timeline frame rate based on this value. This setting is unrelated to capture or transport speeds.

When using the optional Audio and KeyCode Reader accessory to scan audio, the reader will automatically adjust for frame rate to maintain an overall sample rate of 48kHz. Timecode output is supported for 16, 24, 25, and 30 fps, and for other frame rates no timecode signal is outputted.

- **File Name Prefix:** Prefix to help identify the scan. This can be the name of your project, such as the title of the film you are scanning.
- **Timestamp Prefix:** Select this checkbox to prefix your scans with a timestamp as well as the 'file name prefix' you specified. Your clips will be saved to independent sub-folders in the destination folder. This checkbox is selected by default.

If you want to save all your clips together in one master destination folder without the timecode in the file name, simply deselect the checkbox.

NOTE: If you don't make capture names unique with the timestamp prefix and the files go into the same location, this could potentially overwrite files.

- **Roll/Card, Reel Number, Clip Number, and Program Name:** These are ways to identify the clip with metadata.
- **Flags:** You can use these color coded flags to tag clips.
- **Good Take:** Corresponds to 'circled take' metadata in the media pool.
- **Log Clip:** Adds a clip to the media pool. After you mark 'in' and 'out' points for a section you want to scan, confirm the metadata is correct, and then click 'log clip'. For more information, refer to the 'Logging and Capturing Individual Clips' and 'Logging and Capturing Multiple Clips' sections in the DaVinci Resolve manual.
- **Batch Clip, Capture Clip, Capture Now, and  Snapshot:** These scanning buttons offer different methods to capture clips. For more information about scanning buttons, refer to the 'Scanning One or More Sections of Film' section of this manual.

Film Scanning Workflows

The following sections describe how to scan film using DaVinci Resolve and to control the Cintel scanner. Throughout, the features outlined in the previous section are presented in the order in which you'll perform each step of the scanning process.

Before You Begin

Before turning your scanner on and loading film, you should first dust the gate to make sure your scans are as clean as possible. This can be accomplished using compressed air, but if the gate is extremely dirty, you can remove it to give it a more thorough cleaning. Once that's finished, turn on the Cintel Scanner, open DaVinci Resolve and create the project you'll be using to scan film, and then click the 'Cintel scan' button on the media page. Now click the 'Film Scanner' tab to select DaVinci Resolve's film scanner panel.

Before you load film into the scanner or do anything else, click the 'calibrate' button at the bottom left of the film scanner panel. While you should always dust the gate of the scanner before loading a new reel of film, clicking the calibrate button eliminates any unremovable blemishes in your scanner's optics from the scans you're about to make.

Load and Align the Film

Load the film you want to scan. In the presence of an image the scanner will automatically align a frame. You should note that the image may be framed incorrectly if you first load blank film leader.

Next, choose the film type. If necessary, use the 'perf nudge' and 'frame' buttons to manually improve the alignment of the framing bar to the scanner's sensor such that the bottom of the previous frame and the top of the next frame are just visible at the top and bottom of the viewer, and the current frame is centered vertically. It's important to make sure the image in the viewer is not zoomed in when you do this.

Focus the Scanner

Just as you need to focus the lens on a camera, you'll need to focus the projected film image on your scanner's sensor. To achieve perfect focus, turn on the Focus Assist checkbox in the Film Scanner capture settings of DaVinci Resolve. This superimposes a focus peaking overlay over the Ultra HD image that's output from the scanner's HDMI output, and is also displayed in DaVinci Resolve's capture window. For the best results, connect an Ultra HD display to your Cintel scanner so that you can monitor at the maximum available resolution while you focus.

With Focus Assist turned on, focus peaking will detect the film grain of the scanned image whenever the film plane is in perfect focus. This enables the operator to focus the scanner even if the film image is out of focus. Simply monitor the Ultra HD output of the scanner while you turn the Cintel scanner's focus wheel. Your image will be in focus when the grain running throughout the image displays peaking outlines.

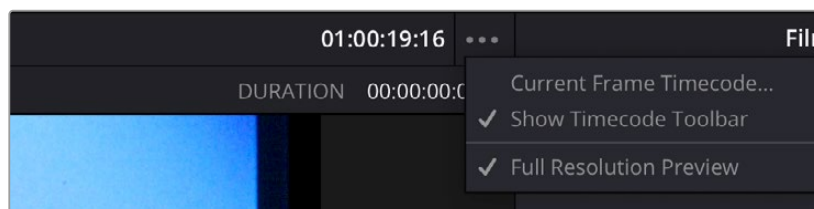
You can verify the focal adjustments you've made by checking the edges of your film's perforations. When these are sharp, your film will be in focus.

Reset the Timecode

To set the timecode for the roll of film you're about to scan, you need to locate the zero frame for that roll. It's standard practice to punch a small physical hole within the frame before the first frame of necessary film on a roll, to use as a permanent reference for whenever that roll is scanned. This is referred to as the marker frame, lab roll hole, or head punch. By always setting the first frame of timecode to match the marker frame, subsequent film scans will have the same frame count as previous scans, making it possible to rescan and reconform the same material whenever necessary.

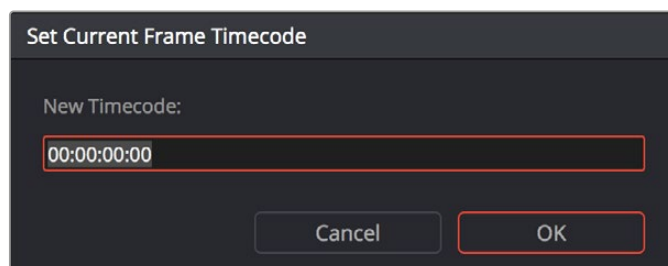
To reset scanned timecode at the marker frame of a new film roll:

- 1 Use the transport controls under the viewer to locate the marker frame.
- 2 Click the 'viewer' option menu and choose 'current frame timecode.'



Choosing Current Frame Timecode from the Viewer Option menu

- 3 Enter a timecode value in the dialog box that appears. For example, if you're scanning the first roll of a project, you can enter 01:00:00:00.



The Set Current Frame Timecode dialog

- 4 When you're done, click OK.

Timecode cannot be a negative value, so don't set the start frame to zero. Another common organizational technique is to change the hour number whenever you change rolls, to coincide with the film roll's number, which makes it easy to identify a scanned clip with the corresponding source roll and frame range.

Your Cintel Scanner has built in 'Options Interface' ports for adding optional hardware in the future. This offers the ability to add optional features such as reading KeyCode from the camera negative, or optical/magnetic audio. For more information, see the 'Optional Audio and KeyCode Reader' section in the *Blackmagic Cintel manual*.

Choose a Location to Save the Scanned Frames

Once all this is done, scroll down to the 'capture info' controls in DaVinci Resolve's film scanner panel, and click the 'browse' button to choose a location for the scanned files. You can use the other fields in this section to set what prefix you want to add to the name of the scanned files and enclosing folders. The 'file name prefix' updates the file name preview that's shown at the top in the header. The header also shows the file path, resolution, frame rate, duration, and the format. Specify what roll, reel, clip, and program information you want associated with the scanned media. The 'timestamp prefix' checkbox in the 'Capture info' controls is selected by default and will save your clips to independent sub-folders within the destination folder, together with a timecode prefix in the file name.

If you want to save all your clips together in one master destination folder, simply deselect the checkbox.

When you capture an HDR clip, the scanner completes a high exposure scan and saves it in a hidden folder named .HDR inside the same folder as the standard scan. If you delete the .HDR folder, the scan converts to a normal clip after refreshing it in the media storage and re-importing the clip into media pool. This is useful if there is a problem with the HDR portion of the scan, as you can easily convert it to a regular CRI clip.

Check the Codec

DaVinci Resolve selects the 'Cintel Raw' codec by default, or you can choose 'Cintel Raw 3:1'.

The Cintel Raw Format

The Cintel Raw Format Bayer pattern of each film frame scanned with your Cintel scanner's sensor is saved with embedded scanner metadata as a 12-bit linear Cintel Raw Image, or CRI, image sequence. When grading in DaVinci Resolve, CRI images are automatically debayered as 12-bit log encoded image data.

The logarithmic encoding is similar, but not identical to Cineon encoding. For example, negative film is encoded using a Gamma of 2.046 for density, while print film is encoded using a full range Gamma 2.2 curve to ensure that no image data is clipped. Both of these logarithmic encodings can be converted to a linear color space using the 'Cintel to Linear' 1D LUT, before converting to other color spaces you may want to work in.

The film is scanned using the full sensor area of 4096x3072 to keep the audio waveform visible for optical audio and to accommodate perforation visibility for stabilization. The image is then cropped and the resolution of the capture files depends on the source film format after overscan for perforations and the audio area are removed. For more information about scanning resolutions for different types of film, see the 'specifications' section.

The Cintel scanner creates Cintel Raw files with variable bitrate lossless compression by default. This is visually lossless compression and achieves approximately 3:2 reduction in file size depending on image content. However, Cintel Raw 3:1 uses lossy compression with a ratio of approximately 3:1. This is still very high quality but may not always be visually lossless.

For example, files for 35mm 4 perf are approximately 12.5MB with Cintel Raw and approximately 6.3MB with Cintel Raw 3:1. Files for 16mm are approximately 4MB with Cintel Raw and approximately 2MB with Cintel Raw 3:1.

CinemaDNG Quality Settings

To control the quality of CRI files, use the 'decode quality' and 'play quality' CinemaDNG settings located in the Camera Raw panel of the project Settings. These settings are 'full' by default. On computers with low processor or memory resources, these settings may be lowered but this will affect the quality of the final render.

Set the Timeline Resolution

DaVinci Resolve displays and renders the output from the scanner using the same resolution as the timeline. For example, for 35mm 4 perforation film, a custom resolution of 4096x3072 would be required for maximum resolution.

NOTE: If your timeline is set for HDR with the desired deliverable at Ultra HD, a loss of resolution may occur.

For more information on the cropped image area resolutions for all film gauges, refer to the 'effective resolutions' in the 'specifications' section. Alternatively, for the full native resolution of the captured clip, access the 'clips attributes' in DaVinci Resolve.

Adjusting the Color of the Scanner

DaVinci Resolve's film scanner panel gives you control over the exposure and color temperature of the light used to illuminate the film for scanning. You can adjust these via the light source master wheel and RGB controls, in order to maximize the amount of information you're extracting from each frame, while preventing any part of the image from being irretrievably clipped. While it's true that CRI is a raw image format, there's no latitude beyond the internal data range used by DaVinci, so be mindful that if you're clipping data in the built in video scopes while scanning, it might be clipped permanently in the scanned media.

How often you'll adjust the color and exposure of scanned shots depends on how much variety there is in the scenes on a particular film roll. For example, some rolls may have many takes of the same scene, all of which have the same lighting and which can share the same adjustments.

Meanwhile, other rolls may have a variety of different scenes with widely different lighting in each one, necessitating you to make individual adjustments for each scanned clip to maximize data quality.

This is important because the light source master wheel and RGB controls cannot be automatically changed between scanned clips in a log and capture workflow. This means that the current light source settings will be used for all clips you scan until you manually change those settings again, even for clips that you've logged from different parts of a film roll. This means that the log and capture style of working is only advisable in situations where it makes sense to log multiple clips that share the same light source master wheel and RGB control adjustments.

Otherwise, it's recommended you make lighting adjustments on a clip by clip basis, as you scan each clip, in situations where you need maximum image quality for finishing. Keep in mind that the goal for these adjustments is to maximize image data from the scan, not to create the final look of the clips, which you'll accomplish later in the grading phase of work using the controls of the 'color' page.

To adjust the light source settings, find a typical image for the section of roll or for the first series of shots you're going to scan, and adjust the light source while viewing the built in video scopes.

Adjust the light source master wheel to set the intensity of the light source used to illuminate the film, raising or lowering the level of the R, G, and B channels all at once. For a typical camera negative, this lets you adjust the black point of the film image. In a negative print, the darkest part of the image corresponds to the highlights of the film image. Set the light source master wheel to sit just above the typical Dmin value of 95, as measured on the histogram of the video scopes, which guarantees that the highlights won't be clipped by the Cineon LOG conversion that DaVinci uses to debayer the CRI image for grading. For positive film, manually adjust the light source levels so that no part of the highlights or shadows of the signal is being clipped; typically 1000 in 10-bit or 4000 in 12-bit.

You can turn on 'show reference levels' in the waveform, RGB parade, or histogram scopes, and set the 'low' value to indicate the digital Dmin value of 95.

Once that's accomplished, adjust the RGB controls to rebalance all three color channels by varying amounts to alter the color temperature of the light source used to illuminate the film, to produce the most useful, or neutral, color balance in the scanned result.

Scanning One or More Sections of Film

After you've adjusted the light source, it's a good idea to stay organized as you scan each clip by entering all relevant metadata into the metadata editor as you go. The 'capture info' group of metadata fields contains information for defining the file name prefix, roll, reel number, clip number, program name, flags, and whether a particular take is good. If you populate these fields before scanning a clip, that metadata will be written into the clip.

At the bottom of the 'capture info' panel, you will see four buttons for film scanning.

With all of this accomplished, you can scan the film in one of four ways:

- **Capture Now:** Use the capture now button to capture long sections of a reel all at once. Clicking 'capture now' begins scanning near the current frame, ending whenever you click 'stop'.
If 'Enable 2 Pass HDR Scan' is selected, click 'Capture HDR' after the capture has begun to let DaVinci Resolve know you've reached the end of your desired clip so it can now proceed to capture the high exposure pass. If you scan the entire reel without clicking 'Capture HDR', the scanner automatically proceeds with the high intensity scan from where you started it until the end of the reel.
- **Capture Clip:** A more controlled means of scanning specific sections of film. After you've used the transport controls and the In and Out button to define a section of film, clicking 'capture clip' scans that one clip and then stops.
If 'Enable 2 Pass HDR Scan' is selected, the high intensity HDR scan uses the same In and Out points as the initial scan.
- **Batch Clips:** A way you can log multiple clips in advance of scanning them all at once using the current light source settings in DaVinci Resolve's film scanner panel. Log each clip in advance by setting In and Out points for each section of film you want to scan, and click the 'log clip' button to save that frame range as an unscanned clip in the media pool. When you click 'batch clips', all unscanned clips will be scanned one after the other until the job is complete. You can also select one or more unscanned clips, and only the selected clips will be scanned. Furthermore, you can import an EDL that corresponds to a particular film roll, and use the resulting logged clips for scanning.

NOTE: When you click the 'log clip' button, Cintel Scanner applies the same project settings to all clips in the batch, and uses the newest project settings at the time of capture. You are advised to confirm the scanner settings before starting the batch capture.

If 'Enable 2 Pass HDR Scan' is selected, the high intensity HDR scan uses the same sets of In and Out points as the initial batch of scans.

For more information on batch capture workflows, see *Chapter 24, "Ingesting From Tape."*

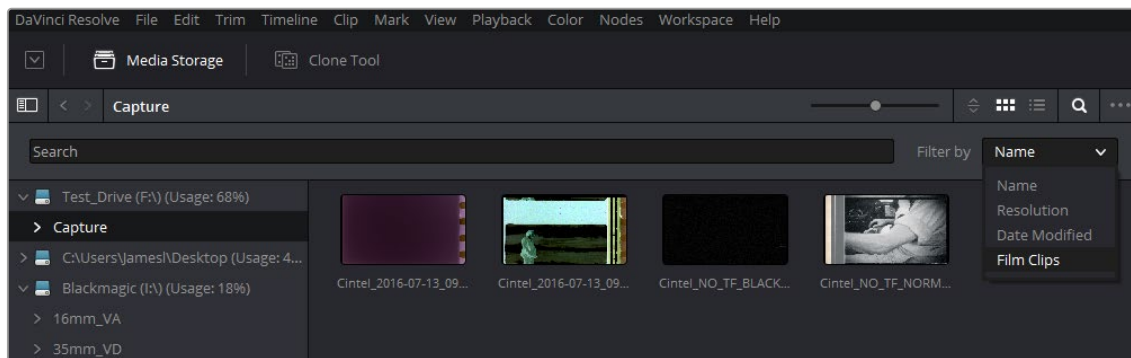
-  **Snapshot:** Capture a single frame with normal exposure and current scanner settings.
Once scanning, if DaVinci Resolve detects that your storage bandwidth is too low to capture at the selected speed, the scan speed will automatically adjust to ensure the capture is successful. If you are using the optional Audio and KeyCode Reader accessory, the audio sample rate will also be adjusted to maintain your chosen audio quality.

Extracting Audio

If the film you're scanning also contains an optical sound track, you can extract the audio in a separate step. There is a standard image frame to audio frame offset of 26 frames for 16mm and 21 frames for 35mm that DaVinci automatically aligns when extracting the audio. Select all of the clips that have an optical sound track, then right-click one of the selected clips and choose 'extract audio'. Resolve analyzes the overlapping optical track area of each frame and automatically generates a matching audio track, synchronized with the scanned image sequence.

Each clip's audio will be automatically extracted, embedded in the clip and saved to the same directory the scanned frames have been written to. A small audio icon will appear on the corner of your clip's thumbnail so you know there is a corresponding audio file.

To make extraction easier, you can filter the clips in the media storage by name, resolution, date modified or by film clips only. Filtering your clips makes it easier for you to find and select exactly what you need. You can also make a large selection and extract audio from multiple clips at once by right clicking on your selection and choosing 'extract audio...' from the menu. During audio extraction, an information box indicates the progress. You can click the 'stop' button any time to stop the extraction.



You can filter the contents in the media storage to make it easier to manage them.

If the 'timestamp prefix' checkbox was deselected in the 'capture info' section when your clips were scanned, and you want to have extracted audio automatically embedded in your clips, always remember to extract audio from the clips inside the media pool.

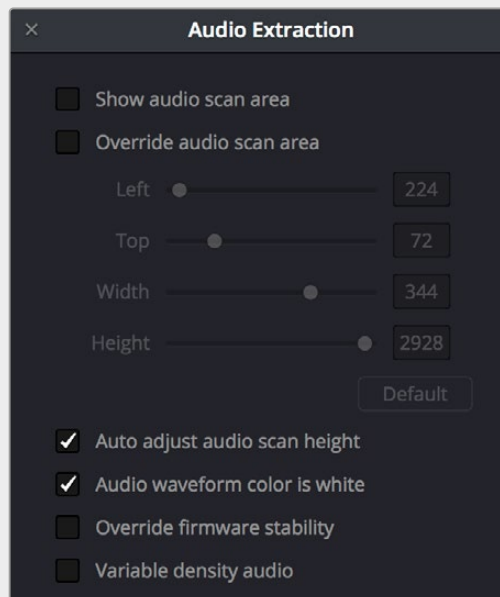
Audio Extraction Settings

Normally, once you have selected the film type, the automatic features in DaVinci Resolve will extract your optical audio perfectly. However, the condition of the optical track can vary with the condition of the film being loaded and in some instances this can confuse the automation. If this happens, you can bypass the automatic features and make adjustments manually.



For manual adjustments, simply open the 'Audio Extraction' settings window by clicking on 'Show Cintel Audio Settings' in the inspector options near the top right of the viewer.

Audio extraction settings let you make the following manual adjustments:



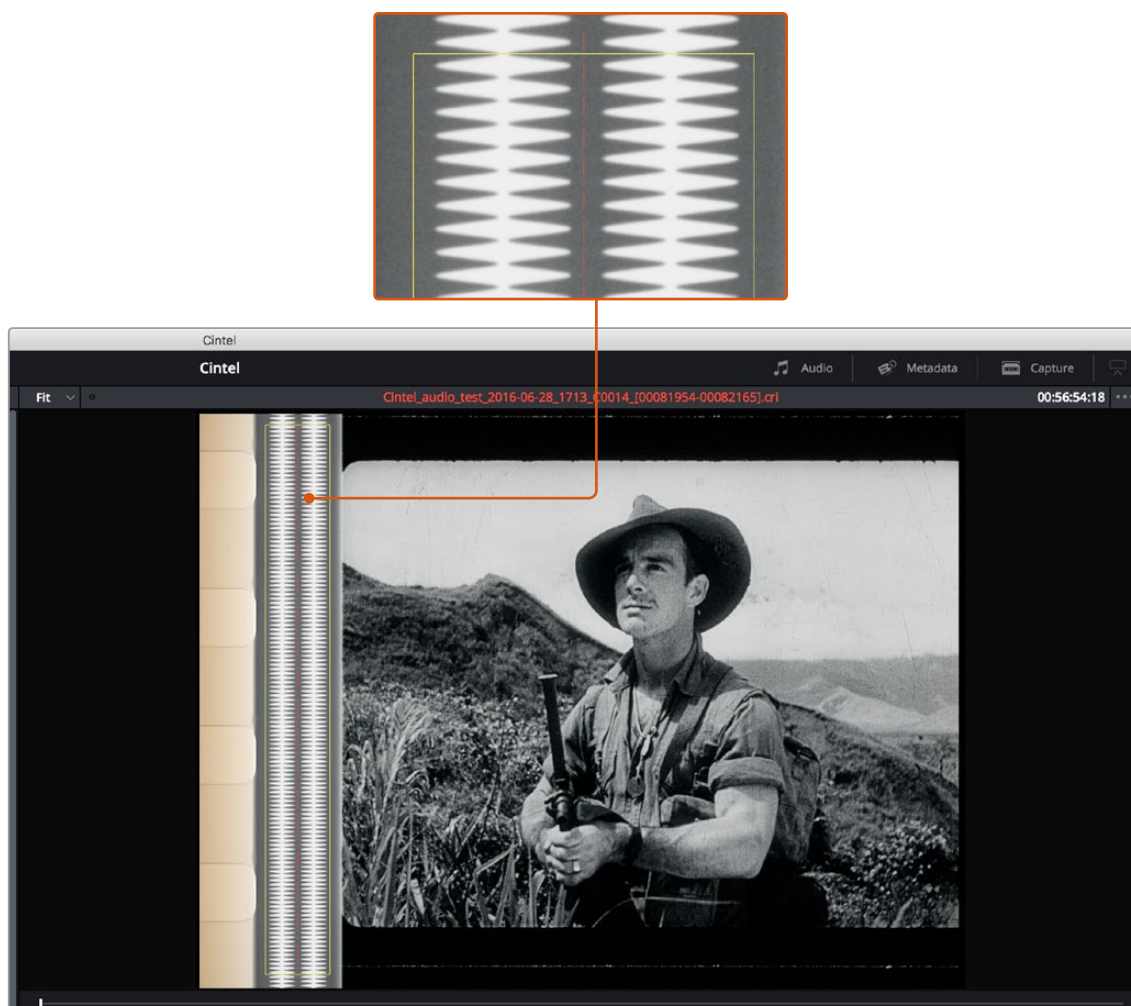
The Audio Extraction settings let you make manual adjustments, if needed.

Show audio scan area

This checkbox turns the audio scan area guides on or off. The guides are displayed as a box on the side of the frame covering the optical audio scan area and shows what optical information will be used during extraction. The position of the guides will conform to the film type you have selected. However, you can change the position manually if you need to. The audio scan area guides are also great indicators to show you what is happening during the extraction process so you can identify any potential troubles and make manual adjustments.

Inside the box is a thin red line. This line is the mid point detector which detects the separation between stereo audio channels. When mono sound is detected during audio extraction, the mid point detector disappears and the guides will adjust automatically to suit the width of the mono optical track.

TIP: If you need a closer inspection of the audio scan area guides, you can zoom into the viewer and move the viewer position up or down, and left or right. Simply choose the amount of zoom from the sizing options at the top left corner of the viewer, then click and drag the viewer with your mouse or track pad.



When 'show audio scan area' setting is turned on, the audio area guides will be visible so you can see exactly what information is being used and monitor the extraction process.

Override audio scan area

This setting provides sliders for adjusting the horizontal and vertical positioning, width, and height of the audio scan area guides.

These settings include:

- **Left and Width:** If your film type is such that audio appears on the right side of the frame, you can simply adjust the 'left' slider to move the guide box to the right. Normally, this will happen automatically if you have the corresponding film type selected, but the setting gives you more flexibility for adjustments if you need it. Similarly, the 'width' setting is used to adjust the width of the scan area.

These are helpful tools for making subtle adjustments to the side edges of the guide box if there are unwanted elements inside the film's optical audio area. This can happen due to perforation wear and tear, or varying print qualities, and can sometimes interfere with the quality of the audio extraction. You can help avoid this by making a subtle movement to the side edges to keep the stray elements outside of the guide box.

- **Top:** This setting adjusts the vertical position of the guide box.
- **Height:** Sometimes film frames on older rolls of film may be slightly smaller than normal due to shrinkage over time. When making manual adjustments to the guide box, you can make adjustments for film shrinkage using the 'height' slider.

- **Auto adjust audio scan height:** This setting is on by default and automatically adjusts the guide box height to align with the audio waveform at the top of each frame. The automatic feature works well for normal audio conditions, however, if during extraction you notice the box moving randomly and the quality of the extraction is affected, it may be due to similar features in the audio track overlapping between frames. If this occurs, deselect the checkbox and try the extraction again. If deselecting the 'Auto adjust audio scan height' checkbox, make sure the 'height' setting places the guide box at the optimal position for the frame. Making manual adjustments can help if you need them, but don't forget to turn the automatic features back on afterwards!

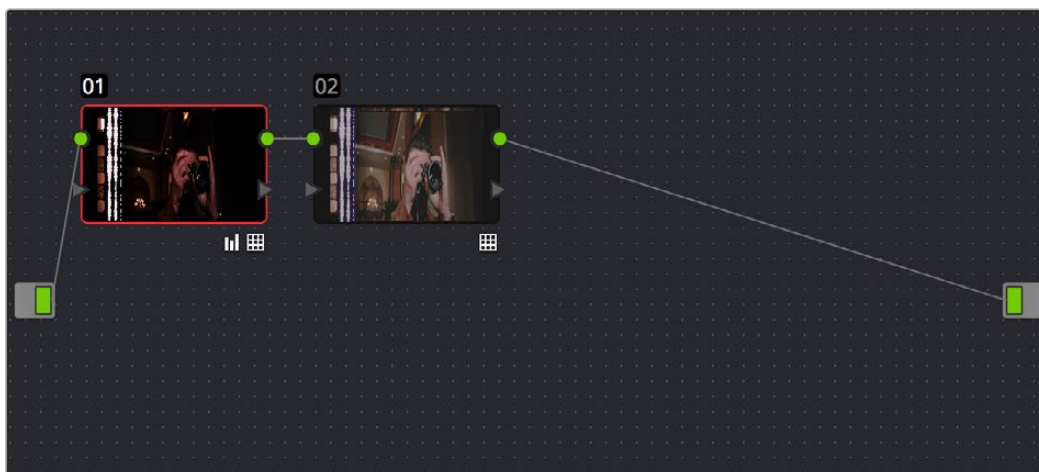
TIP: If deselecting the 'Auto adjust audio scan height' checkbox, make sure the 'height' setting places the guide box at the optimal position for the frame. Making manual adjustments can help if you need them, but don't forget to turn the automatic features back on afterwards!

- **Audio waveform color is white:** Depending on the scanned film type, the audio waveform may be black or white. If the waveform is white, make sure the corresponding checkbox is enabled. This will ensure the white information in the waveform is used during audio extraction. If the waveform is black and the surrounding audio area is white, disable the checkbox so DaVinci knows to use the black information in the waveform. Other automatic features, such as mid point and mono detection, also rely on this setting being set correctly.
- **Override firmware stability:** In rare instances, the condition of the film may have created large movements in the frame due to the internal firmware stabilization. This can cause the audio extraction guide box to misalign with the optical track. If this occurs, enabling 'override firmware stability' lets the audio extraction guide box track the film perforations independently and adjust its positioning for potentially better results.
- **Variable density audio:** If your film contains variable density audio, make sure you select the 'Variable density audio' checkbox so DaVinci Resolve knows the type of audio to extract. The default state is set to 'off' for variable area audio soundtracks. If you haven't used variable density audio before, you can visually identify it as a tight sequence of shaded lines, similar to a bar code with the lines squeezed closer together. By comparison, 'variable area' soundtracks appear as an audio waveform.

Color Space and Sizing

A pair of 1D LUTs, 'Cintel Negative to Linear,' and 'Cintel Print to Linear,' have been provided to help you convert scanned media to a color space in which you can do further work. You can apply these LUTs via a node in the 'color' page to convert the original scans to a Linear color space. However, if you want to convert the image to Rec. 709 or to Cineon for further adjustment, you'll want to apply a second LUT in a second node. The default color space for print is a 2.2 gamma standard log curve, and all others are 2.046 film density log gamma.

In general for negative film, it's best to "color invert" after the second LUT is applied. Furthermore, normally some grading is required on the Linear data to remove black offsets, due to Dmin, for proper conversion into the destination color space. There are a variety of VFX IO LUTs available in the 3D LUT submenu of each node's contextual menu that let you convert an image from Linear color space to any other color space you want to work within.



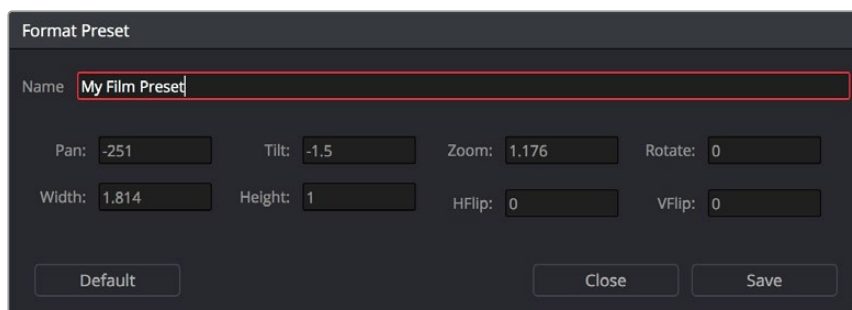
Using three nodes to convert a film scan using LUTs; node 1 converts from Negative or Print to Linear, node 2 converts from Linear to Rec. 709, and node 3, if required, inverts the color

NOTE: Applying a LUT within a node will clip any image data falling below 0 and above 1. To prevent clipping, you can use the Lift/Gamma/Gain controls within any node with a LUT applied to adjust your image levels prior to the transform applied by the LUT within that node.

The format of the film you're scanning and the way the material was originally shot both affect the framing. You can adjust the final framing of your scanned clip by resizing, zooming, stretching, panning, tilting, and more. On the 'Color' page, open the 'Sizing' palette and use the 'Input Sizing' mode to create the necessary framing. To save your sizing preferences as a preset, open the menu, select 'save as new preset' and enter a name for your preset.

Once you've created an appropriate sizing preset for a given type of media, you can apply that preset to multiple film scans all at once, in either the Color page or in the Media Pool using the 'change input sizing preset' command, found in the contextual menu of selected clips.

For more information on sizing, see *Chapter 152, "Sizing and Image Stabilization."*



Creating a sizing preset in the Sizing palette of the Color page



The Cut Page

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